Ian Chi

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Education

- **University of California, Santa Cruz (M.S.) 2025-Present**
  Computer Science Engineering
- **University of California, Davis (B.S.) 2020-2024**
  Physics and Mathematics (high honors), Cumulative GPA:3.859

Skills Summary

- **Programming**:
  - Experienced: Python, Mathematica
  - Other: C, C#, Java, Carnap, Rocq
- **Languages**: English, Chinese (Mandarin)
- **Others**: LaTeX, LT Spice, Lean

Publications & Research

- **Publications & Preprints**:
  1. I. Chi, T. Tan, M. Fraas. **"Poincaré disk as a model of squeezed states of a harmonic oscillator"**.
     Paper Link: https://pubs.aip.org/aip/jmp/article-abstract/66/6/062101/3349274
     Honors Thesis Link: www.math.ucdavis.edu/~webfiles/undergrad_thesis/202403_Ian_Chi_Fraas.pdf
- **Research**
  1. **Simplifying and Generalizing Polylogarithmic-Time Longest Increasing Subsequence (LIS) Algorithms**
     Ongoing work re-examining the Saks & Seshadhri (2010) foundational sublinear-time algorithm. Deconstructing its proofs to improve conceptual clarity, generalize the framework to other DP problems, and optimize constant factors for sublinear LIS estimation.
  2. **"Derivation of Free Flight Distance Distribution with Application to Drift Velocity Calculations"**. I. Chi, R. M. Morin, R. D. Mecholsky, N. A. Mecholsky.
     presented poster available *here*
     Derived probability distributions for free-flight distances and applied statistical models to electron drift velocity calculations within the Drude framework. (Funded by NSF Grant PHY-1852006).
  3. **Laboratory Researcher, DUNE & ANNIE Neutrino Experiments** (UC Davis / Lab)
     Worked with Dr. Robert Svoboda on developing and testing Water Based Slow Scintillators (WBSS) to isolate Cherenkov radiation from scintillation signals. Designed and constructed a double coincidence light yield test bench and performed data analysis on scintillation decay times.

Selected Coursework (UC Santa Cruz, M.S.)

- **Computer Science Engineering**
  - Approximation Algorithms, Complexity Theory, Analysis of Algorithms, Programming Languages, Machine Learning Theory, Modern Algorithms

Selected Coursework (UC Davis, B.S.)

- **Math courses**
  - Real & Complex Analysis, Modern Algebra, Topology, Functional Analysis (Grad), Algebraic Topology (Grad), Probability.
- **Physics courses**:
  - Classical Mechanics, Electricity and Magnetism, Quantum Mechanics, Electronics Lab, Particle Physics, General Relativity, Thermal and Statistical Mechanics, Computational and Mathematical,
- **Philosophy courses**:
  - Symbolic Logic, Philosophy of Mathematics, Philosophy of Science, Philosophy of Sexuality

Awards

- **Citation for Outstanding Performance, UC Davis Mathematics Department**
- **Citation for Outstanding Performance, UC Davis Physics Department**
- **GMTK Game Jam 2023 – Honorable Mention (Top Tier among 6,000+ Entrants)**
  Served as Lead Designer for an independent team developing *SPARED!*, a rapid-prototype puzzle game built under a strict 48-hour deadline. Designed core game mechanics, balancing systems, and player onboarding loops, earning an Honorable Mention out of about 7,000 global entries.
  Portfolio & Playable Builds: `https://splake975.itch.io/`

Extracurricular Activities & Experiences

- **Host of the UCSC TCS Seminar (Fall 2026)** *https://cstheory.ucsc.edu/theory-seminar/*
  Organizing and hosting weekly theory seminars, facilitating discussions on cutting-edge research in theoretical computer science.
- **Applied Probability & Software Project: Advantage Play Modeling** – (Feb 2024 – Dec 2025)
  Designed and implemented software to analyze statistical edges and manage game states for advantage play. Generated \$600k+ in revenue with a 7%+ return over the period. Project concluded following regulatory shifts in California.
- **Nuclear Science and Security Consortium (NSSC) Fellow** (Spring 2023 – Summer 2024)
  Selected for a competitive multidisciplinary fellowship focused on advanced nuclear security and instrumentation.
- **William Lowell Putnam Mathematical Competition** – 14 points (December 2023)
  Achieved a competitive score in North America's premier undergraduate mathematics competition, demonstrating advanced problem-solving skills under tight time constraints.
- **Directed Reading Program – Lie Groups and Lie Algebras** (Spring 2023)
  Departmental reading program under Michael Ragone (PhD Candidate). Studied differential geometry and representation theory via Tu's *Introduction to Manifolds* and Hall's *Lie Groups, Lie Algebras and Representations*.
- **Directed Reading Program – Differential Topology and Geometry** (Fall 2022 – Winter 2023)
  Studied differential forms, co/tangent spaces, and manifolds. Concluded with a departmental presentation under faculty mentorship.

- **Research on Rails Research Trip (Topological Quantum Field Theory)**
  Led by Dr. Andrew Waldron; worked with PhD candidate Petr Vlachopulos studying Michael Atiyah's foundational paper on TQFTs, focusing on the categorical structures of cobordisms and linear maps.